`Assignment 7` Graphical User Interface

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May 29, 2025

GUI

This program is designed as a simplified version inspired by *GeoGebra*. It features a clean interface while offering powerful functionality, integrating CAS (Computer Algebra System), 2D graphing, and 3D calculator capabilities. Input can be made either via keyboard typing or touch-based buttons, supporting both interaction modes. Upon launching the program, it opens in CAS mode by default. You can switch between modes using the menu in the upper right corner—each mode is functionally distinct.

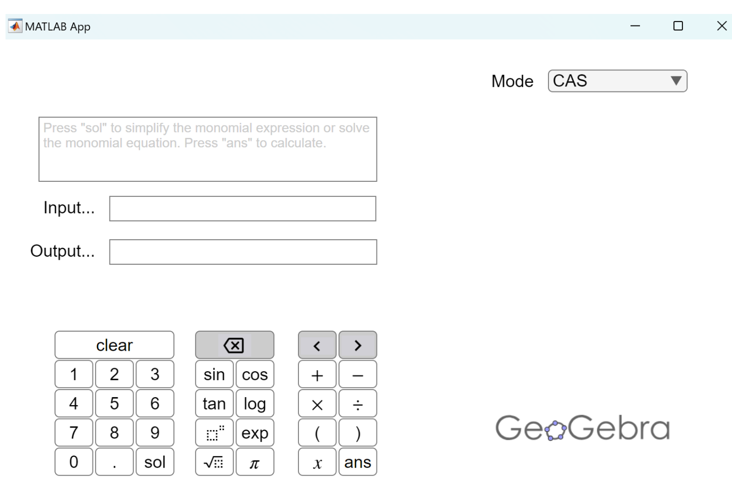

Figure 1: Demo 1

CAS Mode

In CAS mode, you can perform general computations and simplify single-variable expressions (by pressing the `ans` button). For example:

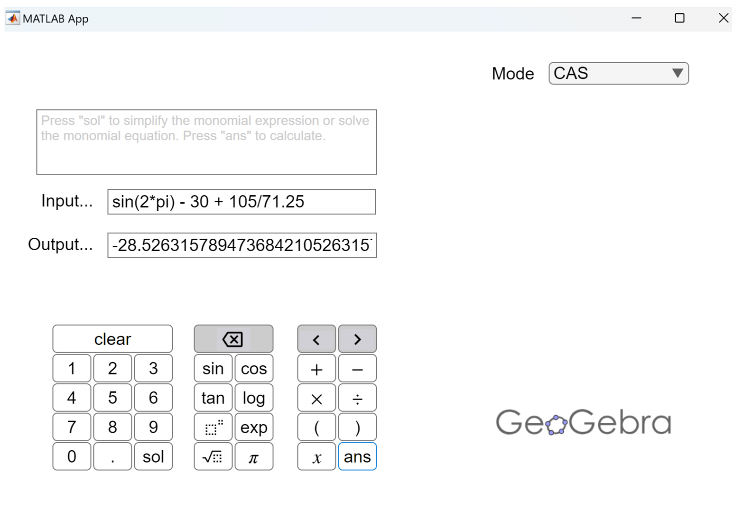

Figure 2: Demo 2

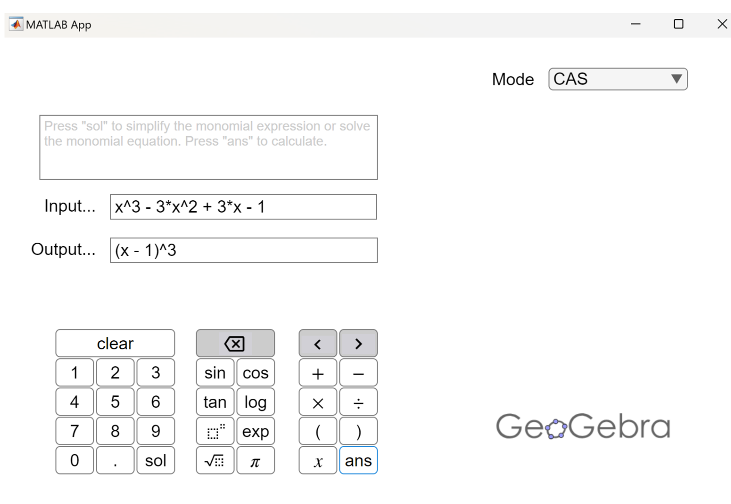

Figure 3: Demo 3

You can also solve a single-variable equation by pressing the `sol` button:

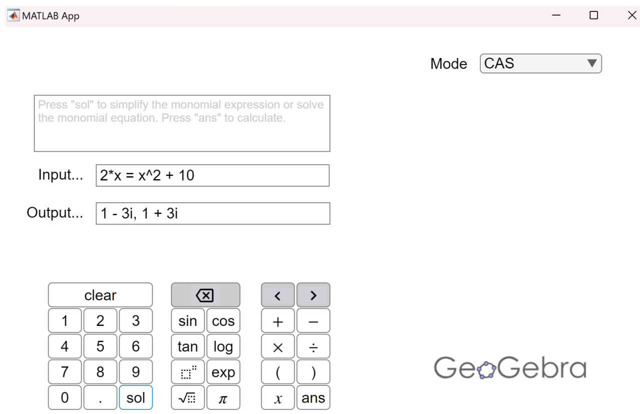

Figure 4: Demo 4

To clear all inputs instantly, simply press the `clear` button.

2D Graphing Mode

In the graphing interface, you can press `add` to include multiple input fields—this is very useful when comparing several function plots simultaneously. If you switch to another mode, remember to press `delete` to remove extra fields! You can also remove any individual field during plotting by clicking `delete`.

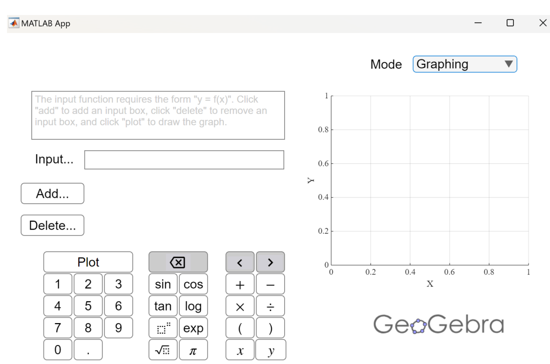

Figure 5: Demo 5

The required input format for 2D graphing is of the form $y = f(x)$:

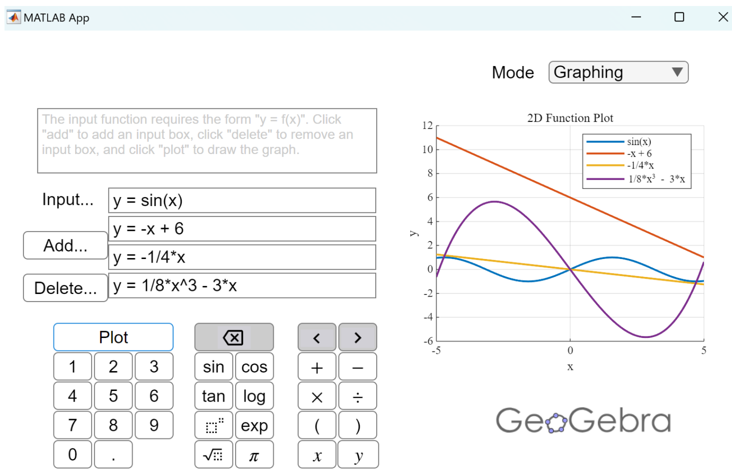

Figure 6: Demo 6

3D Calculator Mode

For 3D plotting, you need to enter functions in the format $z = g(x, y)$:

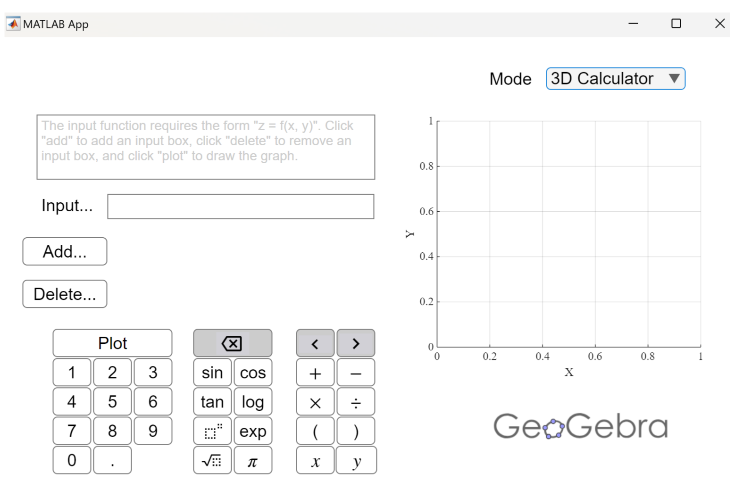

Figure 7: Demo 7

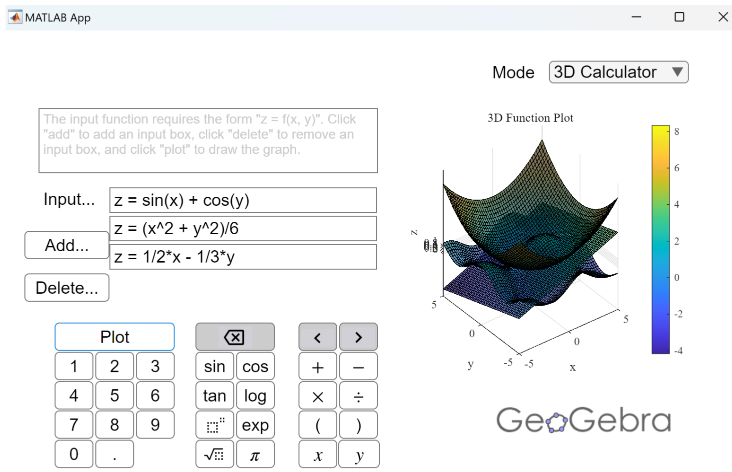

Figure 8: Demo 8

While these input requirement imposes some restrictions on user input flexibility, it improves the robustness and stability of the program to a certain extent.

Although this small app still contains many bugs and lacks some polish in functionality, I want to take this opportunity to pay tribute to the developers of free and open mathematical tools 👏.

This program is not intended for commercial use. If any copyright is violated, please contact me at shepherd.wang@foxmail.com.

```matlab
classdef GeoGebraApp < matlab.apps.AppBase

% Properties that correspond to app components
properties (Access = public)
    UIFigure                matlab.ui.Figure
    TextArea_Graphing_3D    matlab.ui.control.TextArea
    TextArea_Graphing       matlab.ui.control.TextArea
    PlotButton3D            matlab.ui.control.Button
    DeleteButton3D          matlab.ui.control.Button
    DeleteButton2D          matlab.ui.control.Button
    Image                   matlab.ui.control.Image
    ClearButton             matlab.ui.control.Button
    DeleteButton            matlab.ui.control.Button
    RightButton             matlab.ui.control.Button
    LeftButton              matlab.ui.control.Button
    TextArea_CAS            matlab.ui.control.TextArea
    Button_y                matlab.ui.control.Button
    Button_x                matlab.ui.control.Button
    Button_RB               matlab.ui.control.Button
    Button_LB               matlab.ui.control.Button
    DivButton               matlab.ui.control.Button
    MinusButton             matlab.ui.control.Button
    TimesButton             matlab.ui.control.Button
    PlusButton              matlab.ui.control.Button
    piButton                matlab.ui.control.Button
    Button_dot              matlab.ui.control.Button
    Button_0                matlab.ui.control.Button
    Button_9                matlab.ui.control.Button
    Button_6                matlab.ui.control.Button
    Button_3                matlab.ui.control.Button
    Button_8                matlab.ui.control.Button
    Button_5                matlab.ui.control.Button
    Button_2                matlab.ui.control.Button
    Button_7                matlab.ui.control.Button
    Button_4                matlab.ui.control.Button
    Button_1                matlab.ui.control.Button
    AddButton3D             matlab.ui.control.Button
    AddButton2D             matlab.ui.control.Button
    SolveButton             matlab.ui.control.Button
    ModeDropDown            matlab.ui.control.DropDown
    ModeDropDownLabel       matlab.ui.control.Label
    CalculateButton         matlab.ui.control.Button
    PlotButton              matlab.ui.control.Button
    ResultOutput            matlab.ui.control.EditField
    OutputLabel             matlab.ui.control.Label
    ExpressionInput         matlab.ui.control.EditField
    InputLabel              matlab.ui.control.Label
    ExpButton               matlab.ui.control.Button
    LogButton               matlab.ui.control.Button
    TanButton               matlab.ui.control.Button
    CosButton               matlab.ui.control.Button
    SinButton               matlab.ui.control.Button
    SqrtButton              matlab.ui.control.Button
    PowerButton             matlab.ui.control.Button
    UIAxes3D                matlab.ui.control.UIAxes
    UIAxes                  matlab.ui.control.UIAxes
end
```

```

    % Properties for storing input fields and default plotting ...
        settings
    properties (Access = private)
            GraphFields  matlab.ui.control.EditField = ...
                matlab.ui.control.EditField.empty;  % Store ...
                input fields in 2D mode
            Graph3DFields matlab.ui.control.EditField = ...
                matlab.ui.control.EditField.empty; % Store input ...
                fields in 3D mode

            XRange = linspace(-5, 5, 1000);  % Default x range
            XMin = -5;          % Default x minimum
            XMax = 5;           % Default x maximum
            YMin = -5;          % Default y minimum
            YMax = 5;           % Default y maximum
            GridSize = 50;      % Default grid size
        end

        methods (Access = private)

            function updateGraph(app)
                % Try to update the plot; on error, just clear ...
                    the axes
                try
                    app.drawAllFunctions();
                catch
                    cla(app.UIAxes);
                    legend(app.UIAxes, 'off');
                end
            end

            function drawAllFunctions(app)
                % Clear axes and prepare for plotting
                cla(app.UIAxes);
                hold(app.UIAxes, 'on');

                % Combine the main input and any additional fields
                allFields = [app.ExpressionInput, app.GraphFields];

                % Check if there's any valid input
                hasValidInput = false;
                for i = 1:numel(allFields)
                    if ¬isempty(allFields(i).Value)
                        hasValidInput = true;
                        break;
                    end
                end

                if ¬hasValidInput
                    hold(app.UIAxes, 'off');
                    return;
                end

                % Get a set of colors for plotting
                colors = lines(numel(allFields));

```

```
        % Process each input field and plot if valid
        for i = 1:numel(allFields)
            inputStr = allFields(i).Value;

            % Skip empty input
            if isempty(inputStr)
                continue;
            end

            % Extract the expression part after '=' if ...
                present
            if contains(inputStr, '=')
                exprParts = strsplit(inputStr, '=');
                funcStr = strtrim(exprParts{2});
            else
                funcStr = strtrim(inputStr);
            end

            % Remove trailing semicolon if any
            if endsWith(funcStr, ';')
                funcStr = funcStr(1:end-1);
            end

            % Safely compute function values
            x = app.XRange;
            try
                % Create anonymous function and compute y ...
                    values
                fh = str2func(['@(x) ' vectorize(funcStr)]);
                y = fh(x);

                % Plot the curve
                plot(app.UIAxes, x, y, ...
                    'LineWidth', 1.5, ...
                    'Color', colors(i,:), ...
                    'DisplayName', funcStr);
            catch ME
                % Error handling (could add error ...
                    messages here)
            end
        end

        % Set axes properties
        hold(app.UIAxes, 'off');
        grid(app.UIAxes, 'on');
        title(app.UIAxes, '2D Function Plot');
        xlabel(app.UIAxes, 'x');
        ylabel(app.UIAxes, 'y');
    end

    function updateGraph3D(app)
        % Try to update the 3D plot
        try
            app.drawAll3DFunctions();
        catch ME
            % Display error message
            disp(ME.message);
```

```
                cla(app.UIAxes3D);
            end
        end

        function drawAll3DFunctions(app)
            % Clear the 3D axes
            cla(app.UIAxes3D);

            % Combine the main input and any additional 3D ...
                fields
            allFields = [app.ExpressionInput, ...
                app.Graph3DFields];

            % Check if there's any valid input
            hasValidInput = false;
            for i = 1:numel(allFields)
                if ¬isempty(allFields(i).Value)
                    hasValidInput = true;
                    break;
                end
            end

            if ¬hasValidInput
                return;
            end

            % Create a mesh grid for plotting
            x = linspace(app.XMin, app.XMax, app.GridSize);
            y = linspace(app.YMin, app.YMax, app.GridSize);
            [X, Y] = meshgrid(x, y);

            % Get a set of colors for multiple surfaces
            colors = lines(numel(allFields));

            % Process each input field and plot if valid
            for i = 1:numel(allFields)
                inputStr = allFields(i).Value;

                % Skip empty input
                if isempty(inputStr)
                    continue;
                end

                % Extract the expression part after '=' if ...
                    present
                if contains(inputStr, '=')
                    exprParts = strsplit(inputStr, '=');
                    funcStr = strtrim(exprParts{2});
                else
                    funcStr = strtrim(inputStr);
                end

                % Remove trailing semicolon if any
                if endsWith(funcStr, ';')
                    funcStr = funcStr(1:end-1);
                end

```

```
                    % Safely compute function values on the grid
                    try
                        % Create anonymous function and compute z ...
                            values
                        fh = str2func(['@(x,y) ' ...
                            vectorize(funcStr)]);
                        Z = fh(X, Y);

                        % Plot the surface
                        surf(app.UIAxes3D, X, Y, Z, 'FaceAlpha', ...
                            0.7);
                        hold(app.UIAxes3D, 'on');

                    catch ME
                        % Error handling (display message)
                        disp(['Error in function: ' funcStr]);
                        disp(ME.message);
                    end
                end

                % Set axes properties for the 3D plot
                hold(app.UIAxes3D, 'off');
                grid(app.UIAxes3D, 'on');
                title(app.UIAxes3D, '3D Function Plot');
                xlabel(app.UIAxes3D, 'x');
                ylabel(app.UIAxes3D, 'y');
                zlabel(app.UIAxes3D, 'z');
                colorbar(app.UIAxes3D);

                % Add lighting for better visualization
                light(app.UIAxes3D, 'Position', [1 1 1], 'Style', ...
                    'infinite');
                lighting(app.UIAxes3D, 'gouraud');
                material(app.UIAxes3D, 'dull');

                % Set view angle
                view(app.UIAxes3D, 3);
            end
        end

    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            app.ModeDropDown.Value = 'CAS';
            app.ModeDropDownValueChanged(struct('Value','CAS'));
        end

        % Button pushed function: SinButton
        function SinButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'sin('];
        end

        % Button pushed function: CosButton
        function CosButtonPushed(app, event)
```

```
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'cos('];
        end

        % Button pushed function: TanButton
        function TanButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'tan('];
        end

        % Button pushed function: LogButton
        function LogButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'log('];
        end

        % Button pushed function: ExpButton
        function ExpButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'exp('];
        end

        % Button pushed function: SqrtButton
        function SqrtButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'sqrt('];
        end

        % Button pushed function: PowerButton
        function PowerButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '^'];
        end

        % Button pushed function: CalculateButton
        function CalculateButtonPushed(app, event)
            inputStr = app.ExpressionInput.Value;
            % If the string contains "=", take the right-hand ...
                side as the expression
            eqIdx = strfind(inputStr, '=');
            if ¬isempty(eqIdx)
                exprStr = strtrim(inputStr(eqIdx(end)+1:end));
            else
                exprStr = inputStr;
            end
            try
                % Convert the input string to a symbolic expression
                symExpr = str2sym(exprStr);
                simplified = simplify(symExpr); % Perform ...
                    algebraic simplification
                app.ResultOutput.Value = char(simplified); % ...
                    Display the result
            catch
                app.ResultOutput.Value = 'Invalid';
            end
        end

```

```
        % Button pushed function: PlotButton
        function PlotButtonPushed(app, event)
           app.drawAllFunctions();
        end

        % Value changed function: ModeDropDown
        function ModeDropDownValueChanged(app, event)
            mode = app.ModeDropDown.Value;
            switch mode
                case 'CAS'
                    % Clear all the plots
                    cla(app.UIAxes);
                    cla(app.UIAxes3D);
                    app.UIAxes.Visible = 'off';
                    app.UIAxes3D.Visible = 'off';
                    app.ExpressionInput.Visible = 'on';
                    app.ResultOutput.Visible = 'on';
                    app.CalculateButton.Visible = 'on';
                    app.SolveButton.Visible = 'on';
                    app.Button_y.Visible = 'off';
                    app.PlotButton.Visible = 'off';
                    app.AddButton2D.Visible = 'off';
                    app.AddButton3D.Visible = 'off';
                    app.TextArea_CAS.Visible = 'on';
                    app.TextArea_Graphing.Visible = 'off';
                    app.TextArea_Graphing_3D.Visible = 'off';
                    app.ClearButton.Visible = 'on';
                    app.OutputLabel.Visible = 'on';
                    app.DeleteButton2D.Visible = 'off';
                    app.DeleteButton3D.Visible = 'off';
                    app.PlotButton3D.Visible = 'off';

                case 'Graphing'
                    % Clear 2D plot
                    cla(app.UIAxes3D);
                    app.UIAxes.Visible = 'on';
                    app.UIAxes3D.Visible = 'off';
                    app.ExpressionInput.Visible = 'on';
                    app.ResultOutput.Visible = 'off';
                    app.CalculateButton.Visible = 'off';
                    app.SolveButton.Visible = 'off';
                    app.Button_y.Visible = 'on';
                    app.PlotButton.Visible = 'on';
                    app.AddButton2D.Visible = 'on';
                    app.AddButton3D.Visible = 'off';
                    app.TextArea_CAS.Visible = 'off';
                    app.TextArea_Graphing.Visible = 'on';
                    app.TextArea_Graphing_3D.Visible = 'off';
                    app.ClearButton.Visible = 'off';
                    app.OutputLabel.Visible = 'off';
                    app.DeleteButton2D.Visible = 'on';
                    app.DeleteButton3D.Visible = 'off';
                    app.PlotButton3D.Visible = 'off';
                    % Force update the 2D plot
                    app.updateGraph();

                case '3D Calculator'
```

```
                    % Clear 3D plot
                    cla(app.UIAxes);
                    app.UIAxes.Visible = 'off';
                    app.UIAxes3D.Visible = 'on';
                    app.ExpressionInput.Visible = 'on';
                    app.ResultOutput.Visible = 'off';
                    app.CalculateButton.Visible = 'off';
                    app.SolveButton.Visible = 'off';
                    app.Button_y.Visible = 'on';
                    app.PlotButton.Visible = 'off';
                    app.AddButton2D.Visible = 'off';
                    app.AddButton3D.Visible = 'on';
                    app.TextArea_CAS.Visible = 'off';
                    app.TextArea_Graphing.Visible = 'off';
                    app.TextArea_Graphing_3D.Visible = 'on';
                    app.ClearButton.Visible = 'off';
                    app.OutputLabel.Visible = 'off';
                    app.DeleteButton2D.Visible = 'off';
                    app.DeleteButton3D.Visible = 'on';
                    app.PlotButton3D.Visible = 'on';
                    % Force update the 3D plot
                    app.updateGraph3D();
            end
        end

        % Button pushed function: SolveButton
        function SolveButtonPushed(app, event)
            expr = app.ExpressionInput.Value;             % ...
                Expression input
            symEq = str2sym(expr);
            sol = solve(symEq, symvar(symEq));            % The ...
                solution
            app.ResultOutput.Value = join(string(sol), ", ");
        end

        % Button pushed function: AddButton2D
        function AddButton2DPushed(app, event)
            newField = uieditfield(app.UIFigure, 'text', 'Value', ...
                '');
            newField.Position = [115, 284 - ...
                30*numel(app.GraphFields), 283, 27];
            newField.FontSize = 18;
            newField.ValueChangedFcn = @(src,ev) app.updateGraph();
            app.GraphFields = [app.GraphFields, newField];
            app.updateGraph();
        end

        % Button pushed function: AddButton3D
        function AddButton3DPushed(app, event)
            newField = uieditfield(app.UIFigure, 'text', 'Value', ...
                '');
            newField.Position = [115, 284 - ...
                30*numel(app.Graph3DFields), 283, 27];
            newField.FontSize = 18;
            newField.ValueChangedFcn = @(src,ev) ...
                app.updateGraph3D();
            app.Graph3DFields = [app.Graph3DFields, newField];
```

```
            app.updateGraph3D();
        end

        % Button pushed function: Button_1
        function Button_1Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '1'];
        end

        % Button pushed function: Button_2
        function Button_2Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '2'];
        end

        % Button pushed function: Button_3
        function Button_3Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '3'];
        end

        % Button pushed function: Button_5
        function Button_5Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '5'];
        end

        % Button pushed function: Button_9
        function Button_9Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '9'];
        end

        % Button pushed function: Button_4
        function Button_4ButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '4'];
        end

        % Button pushed function: Button_8
        function Button_8Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '8'];
        end

        % Button pushed function: Button_7
        function Button_7Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '7'];
        end

        % Button pushed function: Button_6
        function Button_6Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '6'];
        end

```

```
        % Button pushed function: Button_0
        function Button_0Pushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '0'];
        end

        % Button pushed function: Button_dot
        function Button_dotPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '.'];
        end

        % Button pushed function: piButton
        function piButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'pi'];
        end

        % Button pushed function: PlusButton
        function PlusButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '+'];
        end

        % Button pushed function: MinusButton
        function MinusButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '-'];
        end

        % Button pushed function: TimesButton
        function TimesButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '*'];
        end

        % Button pushed function: DivButton
        function DivButtonPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '/'];
        end

        % Button pushed function: Button_LB
        function Button_LBPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, '('];
        end

        % Button pushed function: Button_RB
        function Button_RBPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, ')'];
        end

        % Button pushed function: Button_x
        function Button_xPushed(app, event)
            app.ExpressionInput.Value = ...
```

```matlab
                    [app.ExpressionInput.Value, 'x'];
        end

        % Button pushed function: Button_y
        function Button_yPushed(app, event)
            app.ExpressionInput.Value = ...
                [app.ExpressionInput.Value, 'y'];
        end

        % Button pushed function: ClearButton
        function ClearButtonPushed(app, event)
            app.ExpressionInput.Value = "";
        end

        % Button pushed function: DeleteButton
        function DeleteButtonPushed(app, event)
            % Read the current text (string or char)
            str = app.ExpressionInput.Value;
            if strlength(str) > 0
                % Remove the last character
                % If 'its a char array, use numel(str); if 'its a ...
                    string, use strlength
                newStr = extractBefore(str, strlength(str));  % ...
                    Remove the last character
                app.ExpressionInput.Value = newStr;
            end
        end

        % Button pushed function: DeleteButton2D
        function DeleteButton2DPushed(app, event)
            % If there are no dynamically generated input fields, ...
                return directly
            if isempty(app.GraphFields)
                return;
            end

            % 1. Delete the last generated input field component
            lastField = app.GraphFields(end);
            delete(lastField);

            % 2. Remove this component handle from the ...
                GraphFields array
            app.GraphFields(end) = [];

            % 3. Rearrange the positions of the remaining input ...
                fields (maintain a gap of 30 pixels)
            %    Assume each input field has a height of 27, ...
                initial y-coordinate is 284, and the gap is 30
            for idx = 1:numel(app.GraphFields)
                newY = 284 - 30*(idx-1);
                app.GraphFields(idx).Position(2) = newY;
            end

            % 4. Call updateGraph() to refresh the graph based on ...
                the modified list of input fields
            app.updateGraph();
        end
```

```matlab

        % Button pushed function: DeleteButton3D
        function DeleteButton3DPushed(app, event)
            % If there are no dynamically generated input fields, ...
                return directly
            if isempty(app.Graph3DFields)
                return;
            end

            % 1. Delete the last generated input field component
            lastField = app.Graph3DFields(end);
            delete(lastField);

            % 2. Remove this component handle from the ...
                Graph3DFields array
            app.Graph3DFields(end) = [];

            % 3. Rearrange the positions of the remaining input ...
                fields (maintain a gap of 30 pixels)
            %    Assume each input field has a height of 27, ...
                initial y-coordinate is 284, and the gap is 30
            for idx = 1:numel(app.Graph3DFields)
                newY = 284 - 30*(idx-1);
                app.Graph3DFields(idx).Position(2) = newY;
            end

            % 4. Call updateGraph3D() to refresh the graph based ...
                on the modified list of input fields
            app.updateGraph3D();
        end

        % Button pushed function: PlotButton3D
        function PlotButton3DPushed(app, event)
            app.updateGraph3D();
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Get the file path for locating images
            pathToMLAPP = fileparts(mfilename('fullpath'));

            % Create UIFigure and hide until all components are ...
                created
            app.UIFigure = uifigure('Visible', 'off');
            app.UIFigure.Color = [1 1 1];
            app.UIFigure.Position = [100 100 787 506];
            app.UIFigure.Name = 'MATLAB App';

            % Create UIAxes
            app.UIAxes = uiaxes(app.UIFigure);
            xlabel(app.UIAxes, 'X')
            ylabel(app.UIAxes, 'Y')
```

```
            zlabel(app.UIAxes, 'Z')
            app.UIAxes.FontName = 'Times New Roman';
            app.UIAxes.XGrid = 'on';
            app.UIAxes.YGrid = 'on';
            app.UIAxes.ZGrid = 'on';
            app.UIAxes.Position = [428 145 321 280];

            % Create UIAxes3D
            app.UIAxes3D = uiaxes(app.UIFigure);
            xlabel(app.UIAxes3D, 'X')
            ylabel(app.UIAxes3D, 'Y')
            zlabel(app.UIAxes3D, 'Z')
            app.UIAxes3D.FontName = 'Times New Roman';
            app.UIAxes3D.ZTick = [0 0.2 0.4 0.6 0.8 1];
            app.UIAxes3D.XGrid = 'on';
            app.UIAxes3D.YGrid = 'on';
            app.UIAxes3D.ZGrid = 'on';
            app.UIAxes3D.Position = [428 145 321 280];

            % Create PowerButton
            app.PowerButton = uibutton(app.UIFigure, 'push');
            app.PowerButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @PowerButtonPushed, true);
            app.PowerButton.Icon = fullfile(pathToMLAPP, ...
                'power_icon.png');
            app.PowerButton.BackgroundColor = [1 1 1];
            app.PowerButton.Position = [206 75 41 30];
            app.PowerButton.Text = '';

            % Create SqrtButton
            app.SqrtButton = uibutton(app.UIFigure, 'push');
            app.SqrtButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @SqrtButtonPushed, true);
            app.SqrtButton.Icon = fullfile(pathToMLAPP, ...
                'sqrt_icon.png');
            app.SqrtButton.BackgroundColor = [1 1 1];
            app.SqrtButton.Position = [206 44 41 30];
            app.SqrtButton.Text = '';

            % Create SinButton
            app.SinButton = uibutton(app.UIFigure, 'push');
            app.SinButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @SinButtonPushed, true);
            app.SinButton.BackgroundColor = [1 1 1];
            app.SinButton.FontSize = 18;
            app.SinButton.Position = [206 137 41 30];
            app.SinButton.Text = 'sin';

            % Create CosButton
            app.CosButton = uibutton(app.UIFigure, 'push');
            app.CosButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @CosButtonPushed, true);
            app.CosButton.BackgroundColor = [1 1 1];
            app.CosButton.FontSize = 18;
            app.CosButton.Position = [249 137 41 30];
            app.CosButton.Text = 'cos';

```

```
            % Create TanButton
            app.TanButton = uibutton(app.UIFigure, 'push');
            app.TanButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @TanButtonPushed, true);
            app.TanButton.BackgroundColor = [1 1 1];
            app.TanButton.FontSize = 18;
            app.TanButton.Position = [206 106 41 30];
            app.TanButton.Text = 'tan';

            % Create LogButton
            app.LogButton = uibutton(app.UIFigure, 'push');
            app.LogButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @LogButtonPushed, true);
            app.LogButton.BackgroundColor = [1 1 1];
            app.LogButton.FontSize = 18;
            app.LogButton.Position = [249 106 41 30];
            app.LogButton.Text = 'log';

            % Create ExpButton
            app.ExpButton = uibutton(app.UIFigure, 'push');
            app.ExpButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @ExpButtonPushed, true);
            app.ExpButton.BackgroundColor = [1 1 1];
            app.ExpButton.FontSize = 18;
            app.ExpButton.Position = [249 75 41 30];
            app.ExpButton.Text = 'exp';

            % Create InputLabel
            app.InputLabel = uilabel(app.UIFigure);
            app.InputLabel.HorizontalAlignment = 'right';
            app.InputLabel.FontName = 'Arial';
            app.InputLabel.FontSize = 18;
            app.InputLabel.Position = [40 318 60 23];
            app.InputLabel.Text = 'Input...';

            % Create ExpressionInput
            app.ExpressionInput = uieditfield(app.UIFigure, 'text');
            app.ExpressionInput.FontSize = 18;
            app.ExpressionInput.Position = [115 314 283 27];

            % Create OutputLabel
            app.OutputLabel = uilabel(app.UIFigure);
            app.OutputLabel.HorizontalAlignment = 'right';
            app.OutputLabel.FontName = 'Arial';
            app.OutputLabel.FontSize = 18;
            app.OutputLabel.Position = [26 272 74 23];
            app.OutputLabel.Text = 'Output...';

            % Create ResultOutput
            app.ResultOutput = uieditfield(app.UIFigure, 'text');
            app.ResultOutput.FontSize = 18;
            app.ResultOutput.Position = [115 268 284 27];

            % Create PlotButton
            app.PlotButton = uibutton(app.UIFigure, 'push');
            app.PlotButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @PlotButtonPushed, true);
```

```
            app.PlotButton.BackgroundColor = [1 1 1];
            app.PlotButton.FontSize = 18;
            app.PlotButton.Position = [57 168 127 30];
            app.PlotButton.Text = 'Plot';

            % Create CalculateButton
            app.CalculateButton = uibutton(app.UIFigure, 'push');
            app.CalculateButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @CalculateButtonPushed, ...
                true);
            app.CalculateButton.BackgroundColor = [1 1 1];
            app.CalculateButton.FontSize = 18;
            app.CalculateButton.Position = [358 44 41 30];
            app.CalculateButton.Text = 'ans';

            % Create ModeDropDownLabel
            app.ModeDropDownLabel = uilabel(app.UIFigure);
            app.ModeDropDownLabel.HorizontalAlignment = 'right';
            app.ModeDropDownLabel.FontName = 'Arial';
            app.ModeDropDownLabel.FontSize = 18;
            app.ModeDropDownLabel.Position = [515 452 50 23];
            app.ModeDropDownLabel.Text = 'Mode';

            % Create ModeDropDown
            app.ModeDropDown = uidropdown(app.UIFigure);
            app.ModeDropDown.Items = {'CAS', 'Graphing', '3D ...
                Calculator'};
            app.ModeDropDown.ValueChangedFcn = ...
                createCallbackFcn(app, @ModeDropDownValueChanged, ...
                true);
            app.ModeDropDown.FontSize = 18;
            app.ModeDropDown.Position = [580 451 148 24];
            app.ModeDropDown.Value = 'CAS';

            % Create SolveButton
            app.SolveButton = uibutton(app.UIFigure, 'push');
            app.SolveButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @SolveButtonPushed, true);
            app.SolveButton.BackgroundColor = [1 1 1];
            app.SolveButton.FontSize = 18;
            app.SolveButton.Position = [143 44 41 30];
            app.SolveButton.Text = 'sol';

            % Create AddButton2D
            app.AddButton2D = uibutton(app.UIFigure, 'push');
            app.AddButton2D.ButtonPushedFcn = ...
                createCallbackFcn(app, @AddButton2DPushed, true);
            app.AddButton2D.BackgroundColor = [1 1 1];
            app.AddButton2D.FontName = 'Arial';
            app.AddButton2D.FontSize = 18;
            app.AddButton2D.Position = [25 265 90 30];
            app.AddButton2D.Text = 'Add...';

            % Create AddButton3D
            app.AddButton3D = uibutton(app.UIFigure, 'push');
            app.AddButton3D.ButtonPushedFcn = ...
                createCallbackFcn(app, @AddButton3DPushed, true);
```

```
            app.AddButton3D.BackgroundColor = [1 1 1];
            app.AddButton3D.FontName = 'Arial';
            app.AddButton3D.FontSize = 18;
            app.AddButton3D.Position = [25 265 90 30];
            app.AddButton3D.Text = 'Add...';

            % Create Button_1
            app.Button_1 = uibutton(app.UIFigure, 'push');
            app.Button_1.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_1Pushed, true);
            app.Button_1.BackgroundColor = [1 1 1];
            app.Button_1.FontSize = 18;
            app.Button_1.Position = [57 137 41 30];
            app.Button_1.Text = '1';

            % Create Button_4
            app.Button_4 = uibutton(app.UIFigure, 'push');
            app.Button_4.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_4ButtonPushed, true);
            app.Button_4.BackgroundColor = [1 1 1];
            app.Button_4.FontSize = 18;
            app.Button_4.Position = [57 106 41 30];
            app.Button_4.Text = '4';

            % Create Button_7
            app.Button_7 = uibutton(app.UIFigure, 'push');
            app.Button_7.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_7Pushed, true);
            app.Button_7.BackgroundColor = [1 1 1];
            app.Button_7.FontSize = 18;
            app.Button_7.Position = [57 75 41 30];
            app.Button_7.Text = '7';

            % Create Button_2
            app.Button_2 = uibutton(app.UIFigure, 'push');
            app.Button_2.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_2Pushed, true);
            app.Button_2.BackgroundColor = [1 1 1];
            app.Button_2.FontSize = 18;
            app.Button_2.Position = [100 137 41 30];
            app.Button_2.Text = '2';

            % Create Button_5
            app.Button_5 = uibutton(app.UIFigure, 'push');
            app.Button_5.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_5Pushed, true);
            app.Button_5.BackgroundColor = [1 1 1];
            app.Button_5.FontSize = 18;
            app.Button_5.Position = [100 106 41 30];
            app.Button_5.Text = '5';

            % Create Button_8
            app.Button_8 = uibutton(app.UIFigure, 'push');
            app.Button_8.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_8Pushed, true);
            app.Button_8.BackgroundColor = [1 1 1];
            app.Button_8.FontSize = 18;
```

```
            app.Button_8.Position = [100 75 41 30];
            app.Button_8.Text = '8';

            % Create Button_3
            app.Button_3 = uibutton(app.UIFigure, 'push');
            app.Button_3.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_3Pushed, true);
            app.Button_3.BackgroundColor = [1 1 1];
            app.Button_3.FontSize = 18;
            app.Button_3.Position = [143 137 41 30];
            app.Button_3.Text = '3';

            % Create Button_6
            app.Button_6 = uibutton(app.UIFigure, 'push');
            app.Button_6.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_6Pushed, true);
            app.Button_6.BackgroundColor = [1 1 1];
            app.Button_6.FontSize = 18;
            app.Button_6.Position = [143 106 41 30];
            app.Button_6.Text = '6';

            % Create Button_9
            app.Button_9 = uibutton(app.UIFigure, 'push');
            app.Button_9.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_9Pushed, true);
            app.Button_9.BackgroundColor = [1 1 1];
            app.Button_9.FontSize = 18;
            app.Button_9.Position = [143 75 41 30];
            app.Button_9.Text = '9';

            % Create Button_0
            app.Button_0 = uibutton(app.UIFigure, 'push');
            app.Button_0.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_0Pushed, true);
            app.Button_0.BackgroundColor = [1 1 1];
            app.Button_0.FontSize = 18;
            app.Button_0.Position = [57 44 41 30];
            app.Button_0.Text = '0';

            % Create Button_dot
            app.Button_dot = uibutton(app.UIFigure, 'push');
            app.Button_dot.ButtonPushedFcn = ...
                createCallbackFcn(app, @Button_dotPushed, true);
            app.Button_dot.BackgroundColor = [1 1 1];
            app.Button_dot.FontSize = 18;
            app.Button_dot.Position = [100 44 41 30];
            app.Button_dot.Text = '.';

            % Create piButton
            app.piButton = uibutton(app.UIFigure, 'push');
            app.piButton.ButtonPushedFcn = createCallbackFcn(app, ...
                @piButtonPushed, true);
            app.piButton.BackgroundColor = [1 1 1];
            app.piButton.FontSize = 18;
            app.piButton.Interpreter = 'latex';
            app.piButton.Position = [249 44 41 30];
            app.piButton.Text = '\pi';
```

```
            % Create PlusButton
            app.PlusButton = uibutton(app.UIFigure, 'push');
            app.PlusButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @PlusButtonPushed, true);
            app.PlusButton.BackgroundColor = [1 1 1];
            app.PlusButton.FontSize = 18;
            app.PlusButton.Interpreter = 'latex';
            app.PlusButton.Position = [315 137 41 30];
            app.PlusButton.Text = '$+$';

            % Create TimesButton
            app.TimesButton = uibutton(app.UIFigure, 'push');
            app.TimesButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @TimesButtonPushed, true);
            app.TimesButton.BackgroundColor = [1 1 1];
            app.TimesButton.FontSize = 18;
            app.TimesButton.Interpreter = 'latex';
            app.TimesButton.Position = [315 106 41 30];
            app.TimesButton.Text = '\times';

            % Create MinusButton
            app.MinusButton = uibutton(app.UIFigure, 'push');
            app.MinusButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @MinusButtonPushed, true);
            app.MinusButton.BackgroundColor = [1 1 1];
            app.MinusButton.FontSize = 18;
            app.MinusButton.Interpreter = 'latex';
            app.MinusButton.Position = [358 137 41 30];
            app.MinusButton.Text = '$-$';

            % Create DivButton
            app.DivButton = uibutton(app.UIFigure, 'push');
            app.DivButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @DivButtonPushed, true);
            app.DivButton.BackgroundColor = [1 1 1];
            app.DivButton.FontSize = 18;
            app.DivButton.Interpreter = 'latex';
            app.DivButton.Position = [358 106 41 30];
            app.DivButton.Text = '\div';

            % Create Button_LB
            app.Button_LB = uibutton(app.UIFigure, 'push');
            app.Button_LB.ButtonPushedFcn = ...
                createCallbackFcn(app, @Button_LBPushed, true);
            app.Button_LB.BackgroundColor = [1 1 1];
            app.Button_LB.FontSize = 18;
            app.Button_LB.Position = [315 75 41 30];
            app.Button_LB.Text = '(';

            % Create Button_RB
            app.Button_RB = uibutton(app.UIFigure, 'push');
            app.Button_RB.ButtonPushedFcn = ...
                createCallbackFcn(app, @Button_RBPushed, true);
            app.Button_RB.BackgroundColor = [1 1 1];
            app.Button_RB.FontSize = 18;
            app.Button_RB.Position = [358 75 41 30];
```

```
            app.Button_RB.Text = ')';

            % Create Button_x
            app.Button_x = uibutton(app.UIFigure, 'push');
            app.Button_x.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_xPushed, true);
            app.Button_x.BackgroundColor = [1 1 1];
            app.Button_x.FontSize = 18;
            app.Button_x.Interpreter = 'latex';
            app.Button_x.Position = [315 44 41 30];
            app.Button_x.Text = '$x$';

            % Create Button_y
            app.Button_y = uibutton(app.UIFigure, 'push');
            app.Button_y.ButtonPushedFcn = createCallbackFcn(app, ...
                @Button_yPushed, true);
            app.Button_y.BackgroundColor = [1 1 1];
            app.Button_y.FontSize = 18;
            app.Button_y.Interpreter = 'latex';
            app.Button_y.Position = [357 44 44 30];
            app.Button_y.Text = '$y$';

            % Create TextArea_CAS
            app.TextArea_CAS = uitextarea(app.UIFigure);
            app.TextArea_CAS.FontSize = 14;
            app.TextArea_CAS.FontColor = [0.8 0.8 0.8];
            app.TextArea_CAS.Position = [40 356 359 69];
            app.TextArea_CAS.Value = {'Press "sol" to simplify ...
                the monomial expression or solve the monomial ...
                equation. Press "ans" to calculate.'};

            % Create LeftButton
            app.LeftButton = uibutton(app.UIFigure, 'push');
            app.LeftButton.Icon = fullfile(pathToMLAPP, ...
                'left_icon.png');
            app.LeftButton.BackgroundColor = [0.8 0.8 0.8];
            app.LeftButton.FontSize = 18;
            app.LeftButton.Interpreter = 'latex';
            app.LeftButton.Position = [315 168 41 30];
            app.LeftButton.Text = '';

            % Create RightButton
            app.RightButton = uibutton(app.UIFigure, 'push');
            app.RightButton.Icon = fullfile(pathToMLAPP, ...
                'right_icon.png');
            app.RightButton.BackgroundColor = [0.8 0.8 0.8];
            app.RightButton.FontSize = 18;
            app.RightButton.Interpreter = 'latex';
            app.RightButton.Position = [358 168 41 30];
            app.RightButton.Text = '';

            % Create DeleteButton
            app.DeleteButton = uibutton(app.UIFigure, 'push');
            app.DeleteButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @DeleteButtonPushed, true);
            app.DeleteButton.Icon = fullfile(pathToMLAPP, ...
                'backspace_icon.png');
```

```
            app.DeleteButton.BackgroundColor = [0.8 0.8 0.8];
            app.DeleteButton.FontSize = 18;
            app.DeleteButton.Interpreter = 'latex';
            app.DeleteButton.Position = [206 168 84 30];
            app.DeleteButton.Text = '';

            % Create ClearButton
            app.ClearButton = uibutton(app.UIFigure, 'push');
            app.ClearButton.ButtonPushedFcn = ...
                createCallbackFcn(app, @ClearButtonPushed, true);
            app.ClearButton.BackgroundColor = [1 1 1];
            app.ClearButton.FontSize = 18;
            app.ClearButton.Position = [57 168 127 30];
            app.ClearButton.Text = 'clear';

            % Create Image
            app.Image = uiimage(app.UIFigure);
            app.Image.Position = [508 44 220 102];
            app.Image.ImageSource = fullfile(pathToMLAPP, ...
                'geogebra.png');

            % Create DeleteButton2D
            app.DeleteButton2D = uibutton(app.UIFigure, 'push');
            app.DeleteButton2D.ButtonPushedFcn = ...
                createCallbackFcn(app, @DeleteButton2DPushed, true);
            app.DeleteButton2D.BackgroundColor = [1 1 1];
            app.DeleteButton2D.FontName = 'Arial';
            app.DeleteButton2D.FontSize = 18;
            app.DeleteButton2D.Position = [25 220 90 30];
            app.DeleteButton2D.Text = 'Delete...';

            % Create DeleteButton3D
            app.DeleteButton3D = uibutton(app.UIFigure, 'push');
            app.DeleteButton3D.ButtonPushedFcn = ...
                createCallbackFcn(app, @DeleteButton3DPushed, true);
            app.DeleteButton3D.BackgroundColor = [1 1 1];
            app.DeleteButton3D.FontName = 'Arial';
            app.DeleteButton3D.FontSize = 18;
            app.DeleteButton3D.Position = [25 220 90 30];
            app.DeleteButton3D.Text = 'Delete...';

            % Create PlotButton3D
            app.PlotButton3D = uibutton(app.UIFigure, 'push');
            app.PlotButton3D.ButtonPushedFcn = ...
                createCallbackFcn(app, @PlotButton3DPushed, true);
            app.PlotButton3D.BackgroundColor = [1 1 1];
            app.PlotButton3D.FontSize = 18;
            app.PlotButton3D.Position = [57 168 127 30];
            app.PlotButton3D.Text = 'Plot';

            % Create TextArea_Graphing
            app.TextArea_Graphing = uitextarea(app.UIFigure);
            app.TextArea_Graphing.FontSize = 14;
            app.TextArea_Graphing.FontColor = [0.8 0.8 0.8];
            app.TextArea_Graphing.Position = [40 356 359 69];
            app.TextArea_Graphing.Value = {'The input function ...
                requires the form "y = f(x)". Click "add" to add ...
```

```
                    an input box, click "delete" to remove an input ...
                    box, and click "plot" to draw the graph.'};

            % Create TextArea_Graphing_3D
            app.TextArea_Graphing_3D = uitextarea(app.UIFigure);
            app.TextArea_Graphing_3D.FontSize = 14;
            app.TextArea_Graphing_3D.FontColor = [0.8 0.8 0.8];
            app.TextArea_Graphing_3D.Position = [40 356 359 69];
            app.TextArea_Graphing_3D.Value = {'The input function ...
                requires the form "z = f(x, y)". Click "add" to ...
                add an input box, click "delete" to remove an ...
                input box, and click "plot" to draw the graph.'};

            % Show the figure after all components are created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = GeoGebraApp

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```